Project Design Phase

Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

Brainstorm & Idea Prioritization – Smart Library System

Introduction

Brainstorming is a creative technique used by teams to generate a wide range of ideas for solving a problem. It encourages open discussion, free thinking, and collaboration among team members. All ideas are welcomed without criticism, allowing innovative and practical solutions.

The main goal is to focus on generating multiple ideas first, then refining and selecting the best solution based on feasibility and impact.

Step-1: Team Gathering, Collaboration and Problem Selection

Our team conducted a brainstorming session where all members discussed challenges faced in traditional library systems.

Problem Statement:

Manual library management processes lead to issues such as:

- Difficulty in tracking book availability
- Time-consuming manual book issue/return process
- Lack of real-time updates
- Poor communication between students and librarians

The team decided to focus on developing a **Smart Library System** to automate and improve library operations.

Step-2: Brainstorm, Idea Listing and Grouping

Ideas Generated:

- Develop an online book search system
- Enable real-time book availability tracking
- Implement online book request and reservation
- Automate book issue and return process
- Send notifications for due dates and approvals
- Create a dashboard for library insights
- Reduce manual record keeping

Grouping of Ideas:

1. Book Management

- Add/Search books
- Track availability
- Update book details

2. User Experience

- Online book request system
- Notification system (email/SMS)
- Easy-to-use interface

3. System Automation

- Automated issue/return process
- Due date tracking
- Fine calculation

4. Analytics & Reporting

- Dashboard visualization
- Reports on issued/overdue books

Step-3: Idea Prioritization

Ideas were prioritized based on feasibility, impact, and ease of implementation.

High Priority Ideas:

- Real-time book availability tracking
- Online book request and issue system
- Automated notifications (due date, approval)

Medium Priority Ideas:

- Dashboard for library insights
- Fine calculation system
- Improved user interface

Low Priority Ideas:

- Advanced analytics and reporting
- AI-based book recommendations

Final Selected Solution

Develop a **Smart Library System** that:

- Provides real-time book availability
- Allows online book request and issue/return
- Sends automatic notifications
- Includes a dashboard for monitoring library activities

This solution improves efficiency, reduces manual work, and enhances user experience.